

Overview

This document is to provide an explanation about how to use the asset registry API using swagger or curl commands. With this, the user will be able to access APIs easily.

Using swagger documentation

The Swagger documentation is available at <https://api-ar.agstack.org/swagger/>. Clicking the link will display the documentation interface.

 **Authentication Guide**

Service Zone	HTTP Method	Credential Type
Authentication	GET/POST	Bearer Token (where applicable)
Field Registration	POST	Bearer Token
Field Retrieval	GET/POST	Bearer Token (where applicable)

 **Quick Start Guide**
1. Login via `/login` to get an `access_token` and `refresh_token`.
2. Use `Bearer_Token` (Lock icon) to authorize protected routes.

Servers

https://api-ar.agstack.org - Production Server

Authorize 

Authentication

Field Registration (POST)

Field Retrieval & Queries

The Asset Registry APIs are protected. To access these APIs, you must log in with your Asset Registry account to obtain an access token. To get the access token follow these steps .

1. To get **access_token** Click on

Authentication

option this will open the login endpoint like this

Authentication

POST /login  User Login

When you click on the endpoint it will take you to the interface where you can access the

Try it out

login api . Just click on and you will be able to enter your login credentials like in the image below.

Get field coordinates using geoid

To get the point from the geoid we have a fetch-field api present in

Field Retrieval & Queries

option. After clicking it will show the apis like this .

Field Retrieval & Queries

- GET /fetch-field/{geo_id} Fetch Field Data
- GET /fetch-field-wkt/{geo_id} Fetch Field WKT
- GET /fetch-field-centroid/{geo_id} Fetch Field Centroid
- POST /get-percentage-overlap-two-fields Get Percentage Overlap

After selecting the option, the available APIs will be displayed as shown below:

Field Retrieval & Queries

- GET /fetch-field/{geo_id} Fetch Field Data
- GET /fetch-field-wkt/{geo_id} Fetch Field WKT
- GET /fetch-field-centroid/{geo_id} Fetch Field Centroid
- POST /get-percentage-overlap-two-fields Get Percentage Overlap

Selecting the **fetch-field** endpoint will then reveal the API access options illustrated in the following image:

GET /fetch-field/{geo_id} Fetch Field Data

Parameters

Name	Description
geo_id * required string (path)	geo_id
s2_index string (query)	s2_index

Responses

Try it out

Just click on and you will be able to enter your **geoid** and **s2_index**(Optional)

GET /fetch-field/{geo_id} Fetch Field Data

Parameters

Name	Description
geo_id * required string (path)	8e7892c80f2593ac03f1d23f79c1e1295b85c1
s2_index string (query)	s2_index

Execute

After entering the credentials hit **Execute**. It will give you Field details in response body Like in the image below.

200

Response body

```
{
  "Geo ID": "8e7892c80f2593ac03f1d23f79c1e1295b85c10940e242a01ec434d93f0f13b6",
  "Geo Data": null,
  "Geo JSON": {
    "geometry": {
      "coordinates": [
        75.8569371700287,
        30.901128143077017
      ],
      "type": "Point"
    },
    "type": "Feature"
  },
  "message": "Field fetched successfully."
}
```

Download

Here you will get the Coordinates in geometry.

Note: GeoJSON follows the standard GIS coordinate order:

- Longitude, Latitude

For example, `75.8569371700287` is the **longitude** and `30.901128143077017` is the **latitude**. Previously, some geoids were incorrectly generated because the coordinate order was reversed, mistakenly treating the first value as latitude and the second as longitude.

Using cURL Commands

Consistent with the Swagger interface, interacting with Asset Registry APIs via cURL requires a valid **access_token** for authentication.

To obtain an access token, authenticate by sending a request to the **/login** endpoint using your registry credentials.

Curl example :

Shell

```
curl -X POST https://user-registry.agstack.org/login \  
  -H "Content-Type: application/json" \  
  -H "X-FROM-ASSET-REGISTRY: True" \  
  -d '{"email": "demo_email@gmail.com", "password": "Demo@12345"}'
```

The API will return a JSON response containing the tokens as shown below:

Response Example :

JSON

```
{  
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",  
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."  
}
```

The **access_token** provided in this response should be used for subsequent API requests.

To retrieve specific coordinates from geoid, execute a request to the **/fetch-field** endpoint.

Curl example :

Shell

```
curl  
https://api-ar.agstack.org/fetch-field/8e7892c80f2593ac03f1d23f79c1  
e1295b85c10940e242a01ec434d93f0f13b6 \  
  -H 'Authorization: Bearer  
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...'
```

Response Example :

```
JSON
{
  "GEO Id":
  "8e7892c80f2593ac03f1d23f79c1e1295b85c10940e242a01ec434d93f0f13b6",
  "Geo Data": null,
  "Geo JSON": {
    "geometry": {
      "coordinates": [
        75.8569371700287,
        30.901128143077017
      ],
      "type": "Point"
    },
    "type": "Feature"
  },
  "message": "Field fetched successfully."
}
```

The response body contains the field coordinates within the geometry object.

Note: GeoJSON follows the standard GIS coordinate order:

- Longitude, Latitude

For example, 75.8569371700287 is the **longitude** and 30.901128143077017 is the **latitude**. Previously, some geoids were incorrectly generated because the coordinate order was reversed, mistakenly treating the first value as latitude and the second as longitude.

This endpoint processes a single geoid per call. To retrieve data for a list of geoids, requests must be executed sequentially (e.g., using a loop).